Assignment 2

Question 1

$x'^{\mu} = \Lambda^{\mu}{}_{\alpha} x^{\alpha}$

$\Lambda^{0}{}_{0} = \gamma = \frac{1}{\sqrt{1-v^2/c^2}}$; $\Lambda^{0}{}_{i} = \frac{1}{\sqrt{1-v^2/c^2}} = \frac{-\gamma v^i}{c} = \Lambda^{i}{}_{0}$; $\Lambda^{i}{}_{j} = \delta^{i}_{j} + \frac{\gamma-1}{v^2} v^i v^j$

Show that these transformation satisfy the condition $\Lambda^{\mu}{}_{\alpha}\, \eta_{\alpha\beta}\, \Lambda^{\nu}{}_{\beta}$

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & \frac{-\gamma v^1}{c} & \frac{-\gamma v^2}{c} & \frac{-\gamma v^3}{c} \\ \frac{-\gamma v^1}{c} & 1 + \frac{\gamma-1}{v^2} v^1 v^1 & \frac{\gamma-1}{v^2} v^1 v^2 & \frac{\gamma-1}{v^2} v^1 v^3 \\ \frac{-\gamma v^2}{c} & \frac{\gamma-1}{v^2} v^2 v^1 & 1 + \frac{\gamma-1}{v^2} v^2 v^2 & \frac{\gamma-1}{v^2} v^2 v^3 \\ \frac{-\gamma v^3}{c} & \frac{\gamma-1}{v^2} v^3 v^1 & \frac{\gamma-1}{v^2} v^3 v^2 & 1 + \frac{\gamma-1}{v^2} v^3 v^3 \end{pmatrix} \quad \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In matrix form this can be written as a $\Lambda^{\mu}{}_{\alpha}\, \eta_{\mu\nu}\, \Lambda^{\nu}{}_{\beta}$ can be written as a : $\Lambda^T \eta \Lambda$

$\Lambda^T \eta \Lambda =$

$$\begin{pmatrix} \gamma & \frac{-\gamma v^1}{c} & \frac{-\gamma v^2}{c} & \frac{-\gamma v^3}{c} \\ \frac{-\gamma v^1}{c} & 1 + \frac{\gamma-1}{v^2} v^1 v^1 & \frac{\gamma-1}{v^2} v^1 v^2 & \frac{\gamma-1}{v^2} v^1 v^3 \\ \frac{-\gamma v^2}{c} & \frac{\gamma-1}{v^2} v^2 v^1 & 1 + \frac{\gamma-1}{v^2} v^2 v^2 & \frac{\gamma-1}{v^2} v^2 v^3 \\ \frac{-\gamma v^3}{c} & \frac{\gamma-1}{v^2} v^3 v^1 & \frac{\gamma-1}{v^2} v^3 v^2 & 1 + \frac{\gamma-1}{v^2} v^3 v^3 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & \frac{-\gamma v^1}{c} & \frac{-\gamma v^2}{c} & \frac{-\gamma v^3}{c} \\ \frac{-\gamma v^1}{c} & 1 + \frac{\gamma-1}{v^2} v^1 v^1 & \frac{\gamma-1}{v^2} v^1 v^2 & \frac{\gamma-1}{v^2} v^1 v^3 \\ \frac{-\gamma v^2}{c} & \frac{\gamma-1}{v^2} v^2 v^1 & 1 + \frac{\gamma-1}{v^2} v^2 v^2 & \frac{\gamma-1}{v^2} v^2 v^3 \\ \frac{-\gamma v^3}{c} & \frac{\gamma-1}{v^2} v^3 v^1 & \frac{\gamma-1}{v^2} v^3 v^2 & 1 + \frac{\gamma-1}{v^2} v^3 v^3 \end{pmatrix}$$

Using mathematica matrix calculation ca be *e*

In[14]:=
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ClearAll[γ]
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$$\text{FullSimplify}\left[\begin{pmatrix} \gamma & \frac{-\gamma\, v1}{c} & \frac{-\gamma\, v2}{c} & \frac{-\gamma\, v3}{c} \\ \frac{-\gamma\, v1}{c} & 1+\frac{\gamma-1}{v^2}\, v1\, v1 & \frac{\gamma-1}{v^2}\, v1\, v2 & \frac{\gamma-1}{v^2}\, v1\, v3 \\ \frac{-\gamma\, v2}{c} & \frac{\gamma-1}{v^2}\, v2\, v1 & 1+\frac{\gamma-1}{v^2}\, v2\, v2 & \frac{\gamma-1}{v^2}\, v2\, v3 \\ \frac{-\gamma\, v3}{c} & \frac{\gamma-1}{v^2}\, v3\, v1 & \frac{\gamma-1}{v^2}\, v3\, v2 & 1+\frac{\gamma-1}{v^2}\, v3\, v3 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \gamma & \frac{-\gamma\, v1}{c} & \frac{-\gamma\, v2}{c} & \frac{-\gamma\, v3}{c} \\ \frac{-\gamma\, v1}{c} & 1+\frac{\gamma-1}{v^2}\, v1\, v1 & \frac{\gamma-1}{v^2}\, v1\, v2 & \frac{\gamma-1}{v^2}\, v1\, v3 \\ \frac{-\gamma\, v2}{c} & \frac{\gamma-1}{v^2}\, v2\, v1 & 1+\frac{\gamma-1}{v^2}\, v2\, v2 & \frac{\gamma-1}{v^2}\, v2\, v3 \\ \frac{-\gamma\, v3}{c} & \frac{\gamma-1}{v^2}\, v3\, v1 & \frac{\gamma-1}{v^2}\, v3\, v2 & 1+\frac{\gamma-1}{v^2}\, v3\, v3 \end{pmatrix}\right] \text{ // MatrixForm}$$

Out[15]//MatrixForm=

$$\begin{pmatrix} \frac{\left(-c^2+v1^2+v2^2+v3^2\right)\gamma^2}{c^2} & -\frac{v1\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & -\frac{v2\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} \\ -\frac{v1\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & 1+v1^2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & v1\,v2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma}{v^4}\right. \\ -\frac{v2\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & v1\,v2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & 1+v2^2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+}{v^4}\right. \\ -\frac{v3\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & v1\,v3\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & v2\,v3\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma}{v^4}\right. \end{pmatrix}$$

Hence

$\Lambda^T\,\eta\,\Lambda\;=$

$$\begin{pmatrix} \frac{\left(-c^2+v1^2+v2^2+v3^2\right)\gamma^2}{c^2} & -\frac{v1\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & -\frac{v2\left(-v^2+v1^2+\right.}{} \\ -\frac{v1\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & 1+v1^2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & v1\,v2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+\right.\right.}{}\right. \\ -\frac{v2\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & v1\,v2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & 1+v2^2\left(\frac{\left(2\,v^2+\left(v1^2+v2^2\right.\right.}{}\right. \\ -\frac{v3\left(-v^2+v1^2+v2^2+v3^2\right)(-1+\gamma)\gamma}{c\,v^2} & v1\,v3\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}\right) & v2\,v3\left(\frac{\left(2\,v^2+\left(v1^2+v2^2+\right.\right.}{}\right. \end{pmatrix}$$

For Simlifying let us calculate

$$\Rightarrow \frac{\left(2\,v^2+\left(v1^2+v2^2+v3^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}=$$

$$\frac{\left(2\,v^2+\left(v^2\right)(-1+\gamma)\right)(-1+\gamma)}{v^4}-\frac{\gamma^2}{c^2}=\frac{(2+(-1+\gamma))(-1+\gamma)}{v^2}-\frac{\gamma^2}{c^2}=\frac{(1+\gamma))(-1+\gamma)}{v^2}-\frac{\gamma^2}{c^2}=\frac{(1+\gamma))(-1+\gamma)}{v^2}-\frac{\gamma^2}{c^2}=$$

$$\frac{\left(1-\gamma^2\right)}{v^2}-\frac{\gamma^2}{c^2}=\frac{c^2\left(1-\gamma^2\right)-v^2\gamma^2}{c^2\,v^2}=\frac{c^2-c^2\gamma^2-v^2\gamma^2}{c^2\,v^2}=\frac{c^2-\left(c^2-v^2\right)\gamma^2}{c^2\,v^2}=\frac{c^2-\frac{c^2\left(c^2-v^2\right)}{\left(c^2-v^2\right)}}{c^2\,v^2}=\frac{c^2-c^2}{c^2\,v^2}=0$$

$$\Rightarrow \frac{\left(-c^2+v^2\right)\gamma^2}{c^2}=-1\left(1-\frac{v^2}{c^2}\right)\gamma^2=-1$$

From these simplification (A) can be writtten as

$$\Lambda^T\,\eta\,\Lambda\;=\;\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Question 2

$$\Lambda(v) = \Lambda(v_1)\,\Lambda(v_2) \qquad \Lambda^{\mu}{}_{\nu}(v) = \begin{pmatrix} \gamma 1 & \frac{-\gamma 1\ \mathrm{v1}}{c} & \frac{-\gamma 1\ \mathrm{v2}}{c} & \frac{-\gamma 1\ \mathrm{v3}}{c} \\ \frac{-\gamma 1\ \mathrm{v1}}{c} & 1+\frac{\gamma 1-1}{v^2}\,\mathrm{v1\,v1} & \frac{\gamma 1-1}{v^2}\,\mathrm{v1\,v2} & \frac{\gamma 1-1}{v^2}\,\mathrm{v1\,v3} \\ \frac{-\gamma 1\ \mathrm{v2}}{c} & \frac{\gamma 1-1}{v^2}\,\mathrm{v2\,v1} & 1+\frac{\gamma 1-1}{v^2}\,\mathrm{v2\,v2} & \frac{\gamma 1-1}{v^2}\,\mathrm{v2\,v3} \\ \frac{-\gamma 1\ \mathrm{v3}}{c} & \frac{\gamma 1-1}{v^2}\,\mathrm{v3\,v1} & \frac{\gamma 1-1}{v^2}\,\mathrm{v3\,v2} & 1+\frac{\gamma 1-1}{v^2}\,\mathrm{v3\,v3} \end{pmatrix}$$

Let us tranform our frame such that two sequential transformation with velocity $\vec{v_1}$ and $\vec{v_2}$ be in the x direction such that $\vec{v_1} = (v_1, 0, 0)$ and $\vec{v_2} = (v_2, 0, 0)$

Hence our Lorentz transformation for such a boost can be written as

$$\Lambda(\vec{v_1}) = \begin{pmatrix} \gamma 1 & \frac{-\gamma 1\ \mathrm{v1}}{c} & 0 & 0 \\ \frac{-\gamma 1\ \mathrm{v1}}{c} & \gamma 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \Lambda\left(\vec{v_2}\right) = \begin{pmatrix} \gamma 2 & \frac{-\gamma 2\ \mathrm{v2}}{c} & 0 & 0 \\ \frac{-\gamma 2\ \mathrm{v2}}{c} & \gamma 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Lambda = \Lambda(\vec{v_1})\,\Lambda\left(\vec{v_2}\right) = \begin{pmatrix} \left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)\gamma 1\,\gamma 2 & -\frac{\gamma 1\ (\mathrm{v1}\,\gamma 1+\mathrm{v2}\,\gamma 2)}{c} & 0 & 0 \\ -\frac{(\mathrm{v1}+\mathrm{v2})\ \gamma 1\,\gamma 2}{c} & \gamma 1\left(\gamma 1+\frac{\mathrm{v1\,v2}\,\gamma 2}{c^2}\right) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

As this is given that $\Lambda(\vec{v_1})\,\Lambda\left(\vec{v_2}\right)$ has to be $\Lambda(\mathrm{v})$ hence $\Lambda^0{}_0 = \gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$

$$\Rightarrow \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)\gamma 1\,\gamma 2$$

$$\Rightarrow \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)\frac{1}{\sqrt{1-\frac{\mathrm{v1}^2}{c^2}}}\,\frac{1}{\sqrt{1-\frac{\mathrm{v2}^2}{c^2}}}$$

$$\Rightarrow \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)\frac{1}{\sqrt{1-\frac{\mathrm{v1}^2}{c^2}}}$$

$$\Rightarrow \sqrt{1-\frac{v^2}{c^2}} = \frac{1}{\left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)\frac{1}{\sqrt{1-\frac{\mathrm{v1}^2}{c^2}}}\,\frac{1}{\sqrt{1-\frac{\mathrm{v2}^2}{c^2}}}}$$

$$\Rightarrow 1-\frac{v^2}{c^2} = \frac{1}{\left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)^2\frac{1}{1-\frac{\mathrm{v1}^2}{c^2}}\,\frac{1}{\sqrt{1-\frac{\mathrm{v2}^2}{c^2}}}}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{\left(1+\frac{\mathrm{v1\,v2}}{c^2}\right)^2\frac{1}{1-\frac{\mathrm{v1}^2}{c^2}}\,\frac{1}{1-\frac{\mathrm{v2}^2}{c^2}}}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{\left(1+\frac{v1\,v2}{c^2}\right)^2 \frac{1}{1-\frac{v1^2}{c^2}} \frac{1}{1-\frac{v2^2}{c^2}}}$$

$$\Rightarrow \frac{v^2}{c^2} = \frac{\left(1+\frac{v1\,v2}{c^2}\right)^2 \frac{1}{1-\frac{v1^2}{c^2}} \frac{1}{1-\frac{v2^2}{c^2}} - 1}{\left(1+\frac{v1\,v2}{c^2}\right)^2 \frac{1}{1-\frac{v1^2}{c^2}} \frac{1}{1-\frac{v2^2}{c^2}}}$$

$$\Rightarrow \frac{v^2}{c^2} = \frac{\left(1+\frac{v1\,v2}{c^2}\right)^2 - \left(1-\frac{v1^2}{c^2}\right)\left(1-\frac{v2^2}{c^2}\right)}{\left(1+\frac{v1\,v2}{c^2}\right)^2}$$

$$\Rightarrow \frac{v^2}{c^2} = \frac{\left(1+\frac{2\,v1\,v2}{c^2}+\left(\frac{v1\,v2}{c^2}\right)^2\right) - \left(1+\left(\frac{v1\,v2}{c^2}\right)^2 - \frac{v1^2}{c^2} - \frac{v2^2}{c^2}\right)}{\left(1+\frac{v1\,v2}{c^2}\right)^2}$$

$$\Rightarrow \frac{v^2}{c^2} = \frac{\left(\frac{v1}{c}+\frac{v2}{c}\right)^2}{\left(1+\frac{v1\,v2}{c^2}\right)^2}$$

$$\Rightarrow \frac{v^2}{c^2} = \left(\frac{\left(\frac{v1}{c}+\frac{v2}{c}\right)^2}{\left(1+\frac{v1\,v2}{c^2}\right)^2}\right)$$

$$\Rightarrow v = \sqrt{\frac{(v1+v2)^2}{\left(1+\frac{v1\,v2}{c^2}\right)^2}}$$

$$\Rightarrow v = \frac{(v1+v2)}{\left(1+\frac{v1\,v2}{c^2}\right)}$$

Question 3

$$\epsilon'_{\mu\nu\lambda\rho} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \frac{\partial x^\gamma}{\partial x'^\lambda} \frac{\partial x^\delta}{\partial x'^\rho} \epsilon_{\alpha\beta\gamma\delta}$$

$\epsilon_{\mu\nu\lambda\rho}$ is antisymmetric under the exchange of any two indices and $\epsilon_{0123} = 1$
Proper Transformation - > Transformation where det(Λ) = +1 and

Improper Transformation - > Transformation where det(Λ) = +1 and

From the notes we know that

$\frac{\partial x^\alpha}{\partial x'^\mu} = \Lambda_\mu{}^\alpha$

Hence $\epsilon'_{\mu\nu\lambda\rho} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\lambda}\frac{\partial x^\delta}{\partial x'^\rho}\epsilon_{\alpha\beta\gamma\delta}$ can be written as a

$\epsilon'_{\mu\nu\lambda\rho} = \Lambda_\mu{}^\alpha \Lambda_\nu{}^\beta \Lambda_\lambda{}^\gamma \Lambda_\rho{}^\delta \epsilon_{\alpha\beta\gamma\delta}$

Which can be seen that above expression is equal to

$\epsilon'_{\mu\nu\lambda\rho} = \det(\Lambda^{-1})\,\epsilon_{\mu\nu\lambda\rho}$ As we know that $\det(A) = \epsilon_{ijk}\, a_{im}\, a_{in}\, a_{io}$

$\epsilon'_{\mu\nu\lambda\rho} = \frac{\epsilon_{\mu\nu\lambda\rho}}{\det(\Lambda)}$

Hence for the Proper tansformation $\det(\Lambda) = +1$ hence $\epsilon'_{\mu\nu\lambda\rho} = \epsilon_{\mu\nu\lambda\rho}$

and for improper transformation $\det(\Lambda) = -1$ hence $\epsilon'_{\mu\nu\lambda\rho} = -\epsilon_{\mu\nu\lambda\rho}$

Question 4

$$A^{\alpha\beta} = \begin{pmatrix} 4 & 1 & 0 & 3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix}$$

$$1)\ A^{\alpha\beta} = A^{\alpha\mu}\eta_{\mu\beta} = \begin{pmatrix} 4 & 1 & 0 & 3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 1 & 0 & 3 \\ -1 & -3 & 0 & 2 \\ -2 & 0 & 2 & 1 \\ -1 & 0 & -2 & 3 \end{pmatrix}$$

$$(2)\quad A_\alpha{}^\beta = \eta_{\mu\alpha} A^{\mu\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.\begin{pmatrix} 4 & 1 & 0 & 3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix} = \begin{pmatrix} -4 & -1 & 0 & -3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix}$$

$$(3)\ A_{\alpha\beta} =$$

$$\eta_{\alpha\mu} A^{\mu\nu} \eta_{\nu\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 4 & 1 & 0 & 3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -4 & -1 & 0 & -3 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & -2 & 3 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 0 & -3 \\ -1 & -3 & 0 & 2 \\ -2 & 0 & 2 & 1 \\ -1 & 0 & -2 & 3 \end{pmatrix}$$

Question 5

$$A^{\mu\nu} = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -2 & 0 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 4 \end{pmatrix}$$

$A_{\mu\nu} = \eta_{\mu\alpha} A^{\alpha\beta} \eta_{\nu\beta}$

a) $A_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -2 & 0 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 4 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$A_{\mu\nu} = \begin{pmatrix} 0 & -1 & 0 & -1 \\ 0 & -2 & 0 & 2 \\ -2 & 3 & 0 & 1 \\ -1 & 2 & -2 & 4 \end{pmatrix}$$

b) If the frame O' moves with the with velocity v1 then the tranformation can be written as

$$\Lambda'^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & \frac{-\gamma v^1}{c} & \frac{-\gamma v^2}{c} & \frac{-\gamma v^3}{c} \\ \frac{-\gamma v^1}{c} & 1 + \frac{\gamma-1}{v^2} v^1 v^1 & \frac{\gamma-1}{v^2} v^1 v^2 & \frac{\gamma-1}{v^2} v^1 v^3 \\ \frac{-\gamma v^2}{c} & \frac{\gamma-1}{v^2} v^2 v^1 & 1 + \frac{\gamma-1}{v^2} v^2 v^2 & \frac{\gamma-1}{v^2} v^2 v^3 \\ \frac{-\gamma v^3}{c} & \frac{\gamma-1}{v^2} v^3 v^1 & \frac{\gamma-1}{v^2} v^3 v^2 & 1 + \frac{\gamma-1}{v^2} v^3 v^3 \end{pmatrix}$$

$$v^1 = v^1,\ v^2 = 0,\ v^3 = 0 \quad \Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & \frac{-\gamma v^1}{c} & 0 & 0 \\ \frac{-\gamma v^1}{c} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{----------} \ (\boldsymbol{B})$$

$A'^{\alpha\beta} = \Lambda^{\alpha}{}_{\mu} A^{\mu\nu} \Lambda_{\nu}{}^{\beta}$

$A'^{\alpha\beta} = \Lambda A \Lambda^T$

Given that $v^1 = 0.3c,\ v^2 = 0,\ v^3 = 0,\ \gamma = \frac{1}{\sqrt{1-0.09}} = 1.04$

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} 1.04 & -0.3 & 0 & 0 \\ -0.3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$\gamma = \frac{1}{\sqrt{1-0.09}} = 1.04$

$A'^{\alpha\beta} = \Lambda A \Lambda^T$

$$A'^{\alpha\beta} = \begin{pmatrix} 1.04 & -0.3 & 0 & 0 \\ -0.3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -2 & 0 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 4 \end{pmatrix} \begin{pmatrix} 1.04 & -0.3 & 0 & 0 \\ -0.3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -0.492 & 1.64 & 0. & 0.44 \\ 0.69 & -2.3 & 0. & 1.7 \\ 1.18 & 2.4 & 0. & 1. \\ 0.44 & 1.7 & -2. & 4. \end{pmatrix}$$

(c) Determine $A'_{\alpha\beta}$

$A'_{\alpha\beta} = \eta_{\alpha\mu} A'^{\mu\nu} \eta_{\nu\beta}$

In the matrix form expression can be written as a

$A'_{\alpha\beta} = \eta A' \eta$

$$A'_{\alpha\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -0.492 & 1.64 & 0 & 0.44 \\ 0.69 & -2.3 & 0 & 1.7 \\ 1.18 & 2.4 & 0 & 1 \\ 0.44 & 1.7 & -2 & 4 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A'_{\alpha\beta} = \begin{pmatrix} -0.492 & -1.64 & 0. & -0.44 \\ -0.69 & -2.3 & 0. & 1.7 \\ -1.18 & 2.4 & 0. & 1. \\ -0.44 & 1.7 & -2. & 4. \end{pmatrix}$$

d) Comparison

$A'_{\alpha\beta} A'^{\alpha\beta} = 33.1268$

$A_{\mu\nu} A^{\mu\nu} = 35$ (These can be seen from the mathematica code below)

Because they are tensor

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In[44]:= Aup = {{-0.492, 1.64, 0, 0.44}, {0.69, -2.3, 0, 1.7}, {1.18, 2.4, 0, 1}, {0.44, 1.7, -2, 4}};

Adown = {{-0.492, -1.64, 0, -0.44},
   {-0.69, -2.3, 0, 1.7}, {-1.18, 2.4, 0, 1}, {-0.44, 1.7, -2, 4}};

ContractedResult = Total[Flatten[Aup * Adown]]

35

In[47]:= Aup = {{0, 1, 0, 1}, {0, -2, 0, 2}, {2, 3, 0, 1}, {1, 2, -2, 4}};

Adown = {{0, -1, 0, -1}, {0, -2, 0, 2}, {-2, 3, 0, 1}, {-1, 2, -2, 4}};
ContractedResult1 = Sum[Aup[[i, j]] * Adown[[i, j]], {i, 1, 4}, {j, 1, 4}]

Out[49]=
35
```

Question 6

$$F^{\mu\nu} = \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix} \qquad \Lambda^{\beta}{}_{\alpha} = \begin{pmatrix} \gamma & 0 & 0 & \frac{-v\gamma}{c} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-v\gamma}{c} & 0 & 0 & \gamma \end{pmatrix}$$

$F'^{\mu\nu} = \Lambda^{\mu}{}_{\alpha} \Lambda^{\nu}{}_{\beta} F^{\alpha\beta}$

In the matrix form this can be written as *a*

$F'^{\mu\nu} = \Lambda^{\mu}{}_{\alpha} F^{\alpha\beta} \Lambda_{\beta}{}^{\nu}$

$\Lambda_{\beta}{}^{\nu} = [\Lambda^{\mu}{}_{\alpha}]^T$

In the matrix form this can be written as $F' = \Lambda F \Lambda^T$

$$F' = \begin{pmatrix} \gamma & 0 & 0 & -v\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -v\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix} \begin{pmatrix} \gamma & 0 & 0 & -v\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -v\gamma & 0 & 0 & \gamma \end{pmatrix}$$

$$= \begin{pmatrix} 0 & (E1 - B2\, v)\, \gamma & (E2 + B1\, v)\, \gamma & E3 \\ -E1\, \gamma + B2\, v\, \gamma & 0 & B3 & -B2\, \gamma + E1\, v\, \gamma \\ -((E2 + B1\, v)\, \gamma) & -B3 & 0 & (B1 + E2\, v)\, \gamma \\ -E3 & (B2 - E1\, v)\, \gamma & -((B1 + E2\, v)\, \gamma) & 0 \end{pmatrix}$$

$$\boldsymbol{E'.B'} = (E1 - B2\, v)\ \gamma\, (B1 + E2\, v)\, \gamma + (E2 + B1\, v)\ \gamma\ (B2 - E1\, v)\, \gamma + E3\, B3$$
$$= (B1\, E1 + B2\, E2 + B3\, E3) = \boldsymbol{E.B}$$

$$\boldsymbol{E'.E' - B'.B'} = (E1 - B2\, v)\ \gamma\, (E1 - B2\, v)\, \gamma + (E2 + B1\, v)\, \gamma\, (E2 + B1\, v)\, \gamma +$$
$$E3\, E3 - (B1 + E2\, v)\, \gamma\, (B1 + E2\, v)\, \gamma - (B2 - E1\, v)\, \gamma\, (B2 - E1\, v)\, \gamma - E3\, E3$$
$$= \gamma^2 \left((E1 - B2\, v)^2 + (E2 + B1\, v)^2 - (B1 + E2\, v)^2 - (B2 - E1\, v)^2 \right) + E3\, E3 - E3\, E3$$
$$= \left(B1^2 + B2^2 - E1^2 - E2^2\right)\left(-1 + v^2\right) \gamma^2 + E3\, E3 - B3\, B3$$
$$= -B1^2 - B2^2 + E1^2 + E2^2 + E3^2 - B3^2$$
$$= \boldsymbol{E^2 - B^2}$$

In[73]:=

```
FullSimplify[
 (E1 - B2 v) γ (E1 - B2 v) γ + (E2 + B1 v) γ (E2 + B1 v) γ + (-E3 (-1 + v^2) γ^2) (-E3 (-1 + v^2) γ^2)]
FullSimplify[
 γ^2 ((E1 - B2 v)^2 + (E2 + B1 v)^2 - (B1 + E2 v)^2 - (B2 - E1 v)^2) + E3 E3 - E3 E3]
```

Out[73]=

$$\gamma^2 \left((E2 + B1\, v)^2 + (E1 - B2\, v)^2 + E3^2 \left(-1 + v^2\right)^2 \gamma^2 \right)$$

Out[74]=

$$\left(B1^2 + B2^2 - E1^2 - E2^2\right) \left(-1 + v^2\right) \gamma^2$$

Out[56]=

$$\frac{1}{c^2 (c - v) (c + v) \left(c^4 \left(-B1^2 - B2^2 - B3^2 + E1^2 + E2^2 + E3^2\right) + c^2 \left(B1^2 + B2^2 + B3^2 - 2\, E1^2 - E2^2 - E3^2\right) v^2 + 2\, B2\, c\, E1\, v^3 - B2^2\, v^4\right)}$$

```
FullSimplify[{{4, 1, 0, 3}, {1, -3, 0, 2}, {2, 0, 2, 1}, {1, 0, -2, 3}} . {{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}] // MatrixForm
```

```
FullSimplify[{{-4, -1, 0, -3}, {1, -3, 0, 2}, {2, 0, 2, 1}, {1, 0, -2, 3}} . {{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}] // MatrixForm
```

Out[30]//MatrixForm=

$$\begin{pmatrix} -4 & 1 & 0 & 3 \\ -1 & -3 & 0 & 2 \\ -2 & 0 & 2 & 1 \\ -1 & 0 & -2 & 3 \end{pmatrix}$$

Out[31]//MatrixForm=

$$\begin{pmatrix} 4 & -1 & 0 & -3 \\ -1 & -3 & 0 & 2 \\ -2 & 0 & 2 & 1 \\ -1 & 0 & -2 & 3 \end{pmatrix}$$

In[33]:=

$$\text{FullSimplify}\left[\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -2 & 0 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 4 \end{pmatrix} . \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\right] \text{ // MatrixForm}$$

In[40]:=

$$\begin{pmatrix} 0 & -1 & 0 & -1 \\ 0 & -2 & 0 & 2 \\ -2 & 3 & 0 & 1 \\ -1 & 2 & -2 & 4 \end{pmatrix}$$

$$\text{FullSimplify}\left[\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \begin{pmatrix} -0.492 & 1.6400000000000001 & 0 & 0.44000000000000006 \\ 0.69 & -2.3 & 0 & 1.7 \\ 1.1800000000000002 & 2.4 & 0 & 1 \\ 0.44000000000000006 & 1.7 & -2 & 4 \end{pmatrix} . \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\right] \text{ // MatrixForm}$$

Out[40]=

{{0, -1, 0, -1}, {0, -2, 0, 2}, {-2, 3, 0, 1}, {-1, 2, -2, 4}}

Out[41]//MatrixForm=

$$\begin{pmatrix} -0.492 & -1.64 & 0. & -0.44 \\ -0.69 & -2.3 & 0. & 1.7 \\ -1.18 & 2.4 & 0. & 1. \\ -0.44 & 1.7 & -2. & 4. \end{pmatrix}$$

$$\text{FullSimplify}\left[\left(\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \begin{pmatrix} -0.492` & 1.6400000000000001` & 0.` & 0.44000000000000006` \\ 0.69` & -2.3 & 0.` & 1.7` \\ 1.1800000000000002` & 2.4` & 0.` & 1.` \\ 0.44000000000000006` & 1.7` & -2.` & 4.` \end{pmatrix} . \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\right]$$

In[43]:=

```
FullSimplify[{{-0.492`, -1.6400000000000001`, 0.`, -0.44000000000000006`},
  {-0.69`, -2.3`, 0.`, 1.7`},
  {-1.1800000000000002`, 2.4`, 0.`, 1.`},
  {-0.44000000000000006`, 1.7`, -2.`, 4.`}}
 {{-0.492`, 1.6400000000000001`, 0.`, 0.44000000000000006`},
  {0.69`, -2.3`, 0.`, 1.7`},
  {1.1800000000000002`, 2.4`, 0.`, 1.`},
  {0.44000000000000006`, 1.7`, -2.`, 4.`}}] // MatrixForm
```

Out[43]//MatrixForm=

$$\begin{pmatrix} 0.242064 & -2.6896 & 0. & -0.1936 \\ -0.4761 & 5.29 & 0. & 2.89 \\ -1.3924 & 5.76 & 0. & 1. \\ -0.1936 & 2.89 & 4. & 16. \end{pmatrix}$$

In[68]:=

```
Clear[γ]
FullSimplify[{{γ, 0, 0, -v γ}, {0, 1, 0, 0}, {0, 0, 1, 0}, {-v γ, 0, 0, γ}} . {{0, E1, E2, E3}, {-E1, 0, B3, -B2}, {-E2, -B3, 0, B1}, {-E3, B2, -B1, 0}} . {{γ, 0, 0, -v γ}, {0, 1, 0, 0}, {0, 0, 1, 0}, {-v γ, 0, 0, γ}}] // MatrixForm
FullSimplify[
 (E1 - B2 v) γ (B1 + E2 v) γ + (E2 + B1 v) γ (B2 - E1 v) γ - E3 (-1 + v^2) γ^2 B3 ]
```

Out[69]//MatrixForm=

$$\begin{pmatrix} 0 & (E1 - B2\,v)\,\gamma & (E2 + B1\,v)\,\gamma & -E3\,(-1 + v^2)\,\gamma^2 \\ -E1\,\gamma + B2\,v\,\gamma & 0 & B3 & -B2\,\gamma + E1\,v\,\gamma \\ -((E2 + B1\,v)\,\gamma) & -B3 & 0 & (B1 + E2\,v)\,\gamma \\ E3\,(-1 + v^2)\,\gamma^2 & (B2 - E1\,v)\,\gamma & -((B1 + E2\,v)\,\gamma) & 0 \end{pmatrix}$$

Out[70]=

$$-\left((B1\,E1 + B2\,E2 + B3\,E3)\,(-1 + v^2)\,\gamma^2\right)$$

In[71]:=

```
FullSimplify[
 (E1 - B2 v) γ (E1 - B2 v) γ + (E2 + B1 v) γ (E2 + B1 v) γ + (-E3 (-1 + v^2) γ^2) (-E3 (-1 + v^2) γ^2)]
```

Out[71]=

$$\gamma^2\left((E2 + B1\,v)^2 + (E1 - B2\,v)^2 + E3^2\,(-1 + v^2)^2\,\gamma^2\right)$$